

APEX INSTITUTE OF TECHNOLOGY & SCIENCES (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

FINAL ASSESSMENT EXAMINATION EXAMINATION • AY 2024-25

Course: Database Management Systems (CS501) | Faculty: Dr. Ravi Kumar

Date: 10 December 2024

Time: 180 Minutes

Max Marks: 50

Instructions: Answer all questions in Part A. Part B contains internal choice.

PART A — FULL SYLLABUS COMPREHENSIVE

Answer all questions. (50 Marks)

1. Derive the equivalence rules for heuristic query optimization in relational algebra. [15 M]
2. Explain multi-granularity locking protocol and intention locks (IS, IX, SIX). [15 M]
3. Compare optimistic concurrency control with timestamp ordering algorithms. [20 M]